Palo Alto Networks – Factory Resetting a PA220 Next Generation Firewall and Managing the Web Graphic User Interface (GUI)

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Purpose

This guide/analysis is meant for personnel who recently acquired a new or used PA-220 Next Generation Firewall and need a guideline on how to reset all previous configurations and set basic security configurations like a username and password. With an explanation of the Web Graphic User Interface (GUI), the user will be able to monitor and analyze basic parts of network management.

Background

The main device where a majority of the commands/configurations are entered is the Firewall. The Firewall is a device that acts as a network security system that monitors and controls incoming and outgoing network traffic based on configurable security rules called policies. It establishes itself as a barrier between a trusted network and an untrusted network, such as your home network and the Internet. These devices are the backbone of a network that acts as a shield from unauthorized access, harmful activities, and potential threats, and can exist as hardware, software, software-as-a-service (SaaS), or public or private(virtual) cloud.

Before there were firewalls for network security, routers were used in the 1980s for security. This was because they already had the function to separate networks causing them to be able to filter packets crossing them. Firewalls can be categorized as either a network-based or a host-based system.

Network-based firewalls are positioned between 2 or more networks, typically between the LAN and WAN, their basic function being to control the flow of data between connected networks.

The LAN stands for local area network and is a computer network that interconnects computers within a limited area such as a house, campus, or building. Its network equipment and interconnections can be locally managed with Ethernet and Wi-Fi being the 2 most common technologies used for LAN.

The WAN on the other hand, is a telecommunications network that extends over a large geographic area compared to LAN. Wide area networks (WAN) are often established with leased telecommunication circuits. Businesses, School districts and Government entities use WAN to relay data to staff, students, clients, buyers and suppliers from various locations across the world. The Internet for example is considered a WAN. For management, WANs instead of local management have usually large groups of personnel in organizations from across the globe who centrally manage the WAN, however it is up to the discretion of the organization that is in control of the WAN. The best way to think about a WAN is that it is a classification for a network that is used to transmit data over long distances and between networks.

The software application that is being used in this analysis to configure the firewall we are using is called PuTTY. PuTTY in a brief summary is a free and open-source terminal emulator, serial console and network file transfer application. It supports several network protocols, including SCP, SSH, Telnet, rlogin, and raw socket connection. It can also connect to a serial port. Originally it had been written for Microsoft Windows but has been integrated into various operating systems. Although it is an open source emulator, updates and security patches are constantly being worked on by its developers, there are some security flaws, however as long as the software is constantly updated there should be no major flaws.

PuTTY is a heavily used software in many Tech companies and going into the specifics PuTTY is used by 34% of the Information Technology and Services sector, 10% of the Computer Software sector, and 6% of the Telecommunications sector. Most of PuTTY customers are in the U.S. making up a staggering 59%, with 10% in India and 7% in Canada. For the companies we can see a trend where larger companies use the software more than smaller companies, with companies that have over 1,000 employees make up 60% of companies that use the software, Small companies with less than 50 employees make up 9% and Medium sized companies make up 21% of the total companies that use PuTTY. Although many people at first are doubtful about the security and integrity of the application 57% of the companies that use PuTTY are worth over a billion dollars in valuation on average.

Specifically in this analysis, we will be using the Serial Connection Type on the "Session" screen. The Serial Connection Type that you select in PuTTY refers to the fact that you are configuring the program to establish a direct connection to a physical serial port on your local computer. This is used for managing network devices, single-board computers, or other hardware that has a console but for this case that device would specifically be a firewall. Fundamentally a serial connection is a physical connection that bypasses all network security measures that were meant to prevent any unwanted access to the device using a special cable. In this case, the serial connection also labeled as enables command-line access to PuTTY as if you were directly connected to it.

Focusing back on the firewall in this analysis we will be using the Palo Alto Next Generation PA-220 Firewall. The PA-220 is designed for small organizations or branch offices and the main features are active/passive, and active/active high availability (HA), passive cooling (no fans) to reduce noise and power consumption, 8 Ethernet ports, and dual power adapters for power redundancy. The PA-220 firewall enables you to secure your organization through advanced visibility and control of applications, users, content.

The device also uses machine learning (ML) as a core part of its threat prevention and security features. ML is also used to: Provide inline signatureless attack prevention for file-based attacks while identifying and immediately stopping never-before-seen phishing attempts; Push zero-delay signatures and instructions for cloud based processes; detect IoT devices and make policy recommendations; Automate policy recommendations that save time and reduce the chance of human error; and finally used to integrate with Palo Alto's Wildfire Service to analyze unknown files for malicious content and distributes prevention techniques to firewalls within seconds.

In terms of the Operating System the PA-220 for this analysis is using PAN-OS 8.1 which was a significant software release for Palo Alto Networks' Next-Generation firewalls. PAN-OS 8.1 also introduced an improved Apple-ID and Wildfire capabilities, streamlined SSL decryption, and enhanced Panorama management support.

Lab Summary

In this analysis, after factory resetting the PA-220 Palo Alto Next Generation Firewall, we presented a step by step guide on how to change the previous password that was configured on the device as well as how to open the Web GUI to manage the network, which included connection times, IP addresses, and the traffic of packets in the network going through the firewall.

Lab Commands

admin@PA-220>*configure*

This is an essential command for any network administrator to understand on the PA-220 Palo Alto Next Generation Firewall. The command is used to enter configuration mode, allowing the administrator to view, modify, and manage the device's settings via the Command Line Interface (CLI) or in this analysis, the terminal emulator PuTTY. The changes made in this mode are stored in a candidate configuration and not in the base running configuration.

admin@PA-220>*show* system *information*

A command line interface command in the PA-220 Next Generation Firewall used to display essential system information about the firewall. This command provides a quick overview of the device's identity and operational status. The command displays the firewall's Hostname, Management IP Address, Serial Number, Software Version, Uptime and Model which is crucial for any network administrator trying to assess and gain basic information on the firewall.

admin@Pa-220>*ipconfig*

A Windows command line command used to display and manage your computer's network settings. It is an essential utility for basic network troubleshooting and for gathering information about your device's configuration in the network. The command displays the device's IPV4 Address, Subnet Mask, Default Gateway among other miscellaneous that is utilized by network administrators.

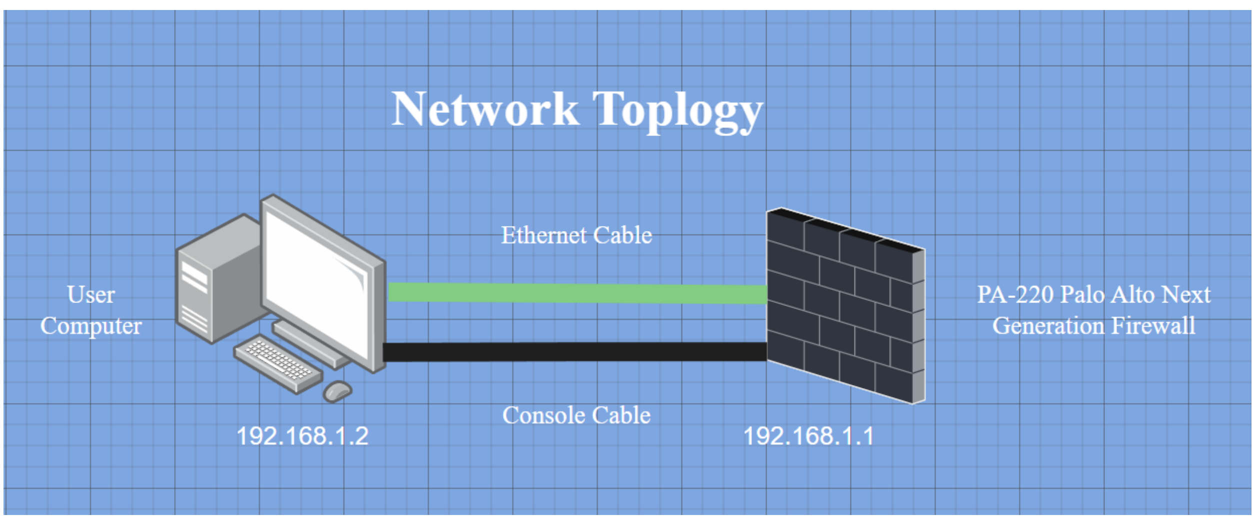
admin@Pa-220>*set mgt-config users <username> password*

This command used in a PA-220 Palo Alto Next Generation Firewall will change the password for a specified user. This command is used after entering configure mode with the *configure* command, and after executing the command there will then be a prompt asking to enter the new password that the network administrator desires to configure.

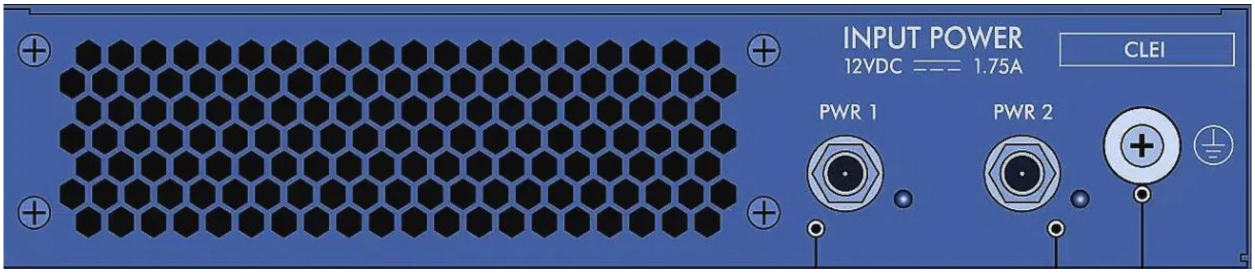
admin@Pa-220>*commit*

Any edits that are done in the GUI or CLI are first saved to temporary staging area called the candidate configuration on the Palo Alto Firewall. The commit command saves and applies configuration changes from the candidate configuration to the running configuration, making them active and enforced. All modifications made in the web interface or command line interface are not active until the commit command is executed. Due to the PA-220 Next Generation Firewall being designed for small offices or small homes the commit speed can take over several minutes to complete.

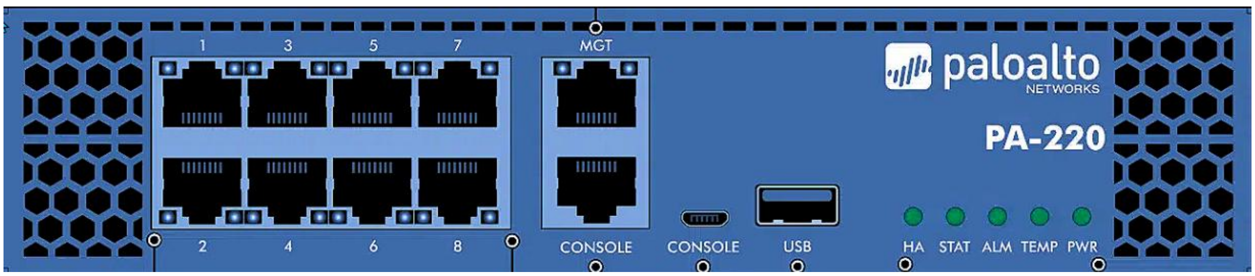
Network Diagram



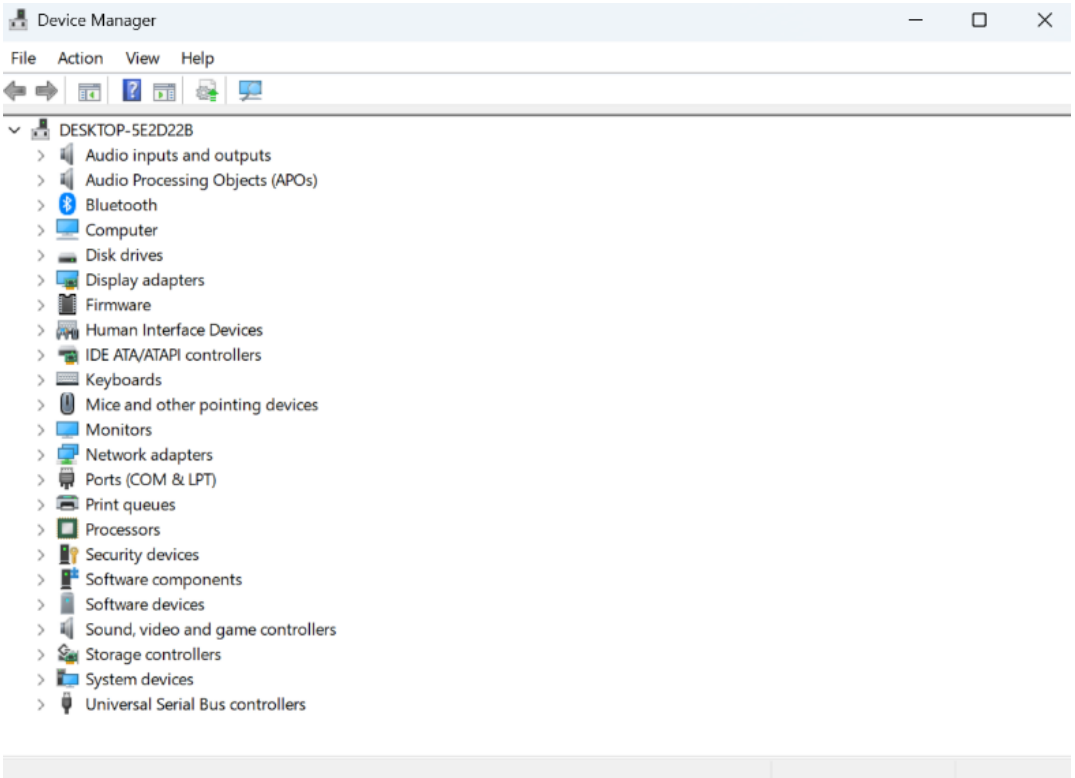
Procedure



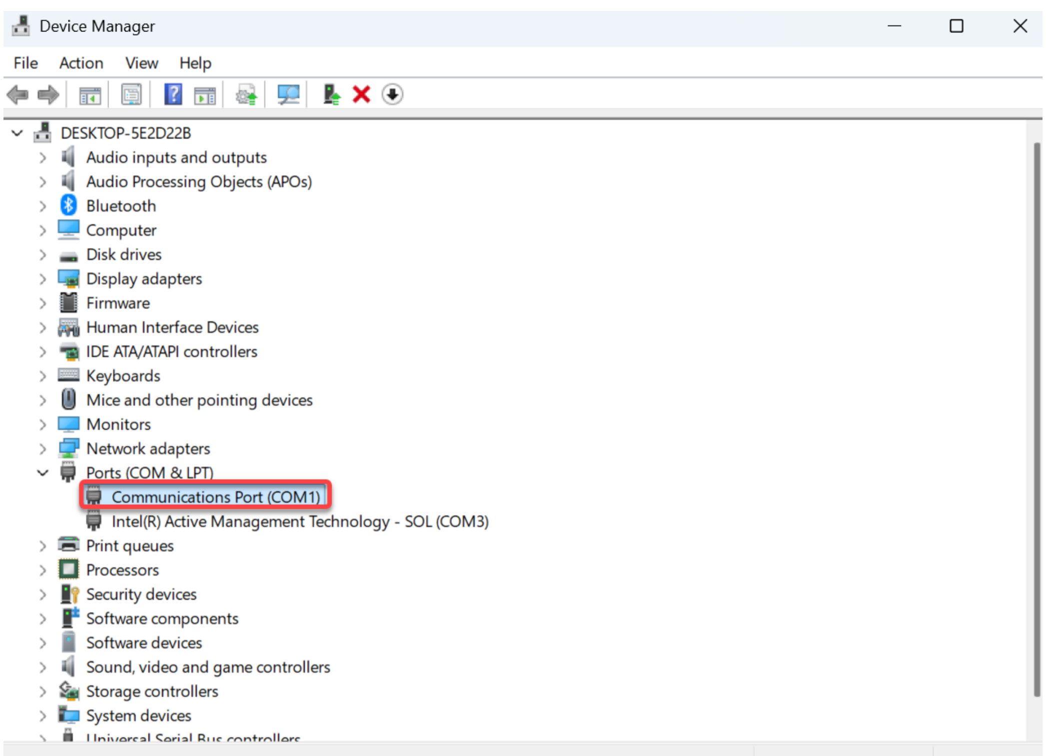
1. Plug in your power cord into either the PWR 1 or 2 port and the other side of the power cord into your computer.



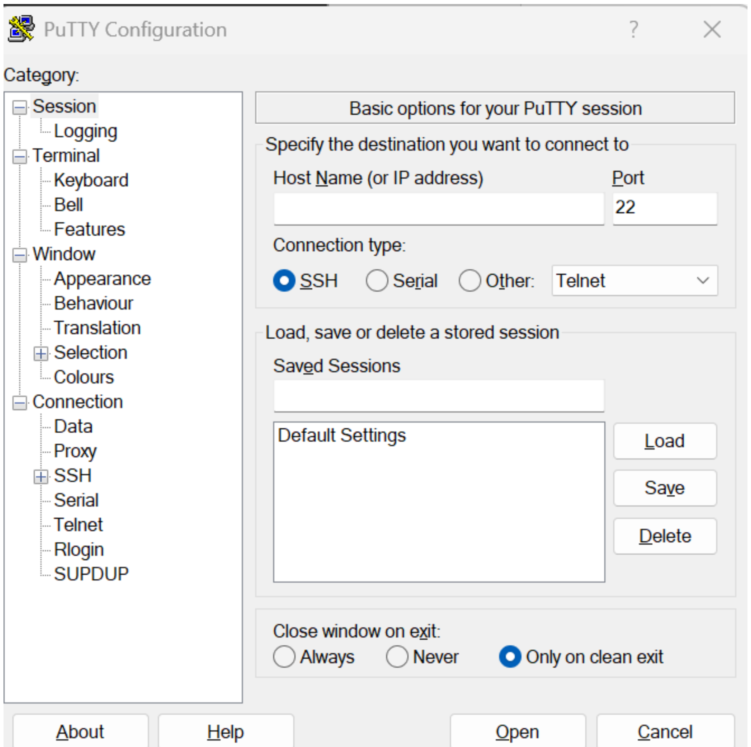
2. Insert your Console Cable into the port labeled Console and the Ethernet Cable into the port labeled MGT with the other side of these cables plugged into your computer.



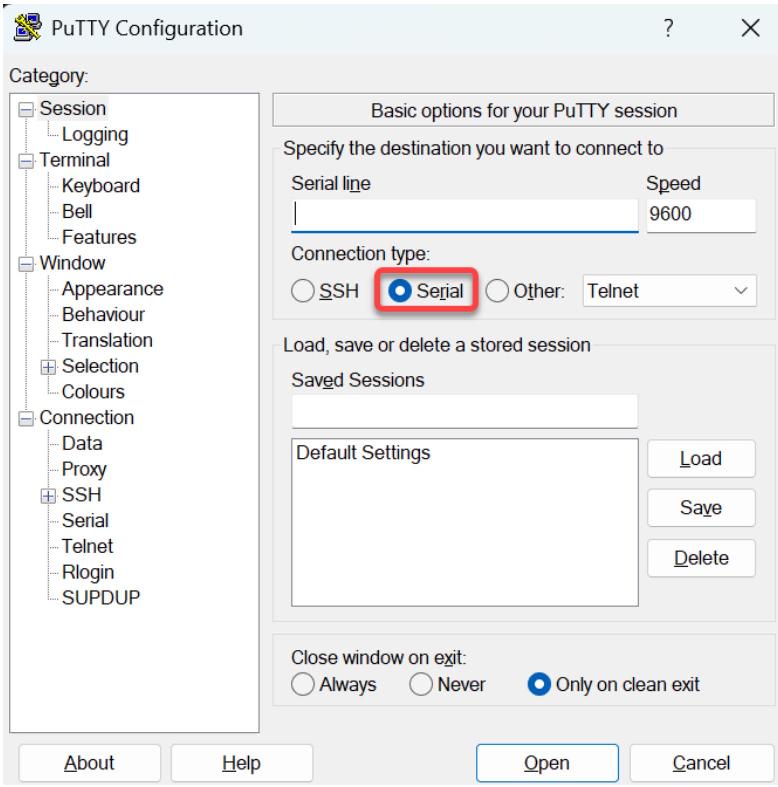
3. Click on the windows button found on the taskbar on the bottom of the screen and type in Device Manager before pressing the enter button. You should be directed to a place screen as shown above.



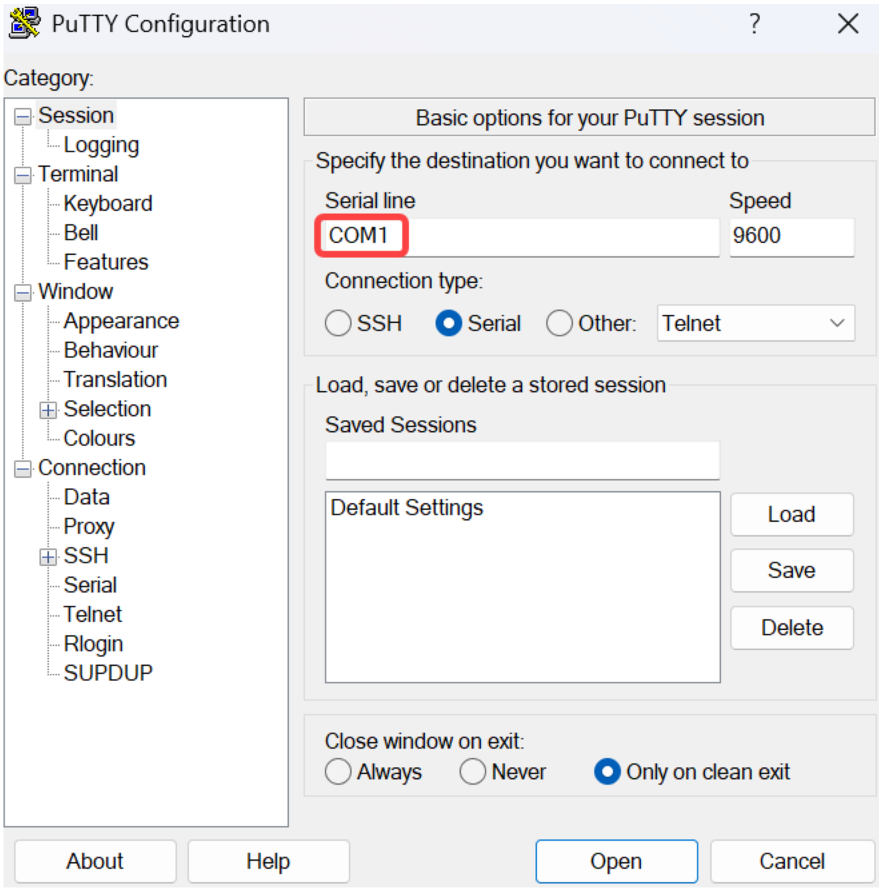
4. Click on the arrow that is to the side of the Ports (COM & LPT) section which shows 2 more sections. Note the specific COM port that is found as highlighted above and in this situation that would be COM1.



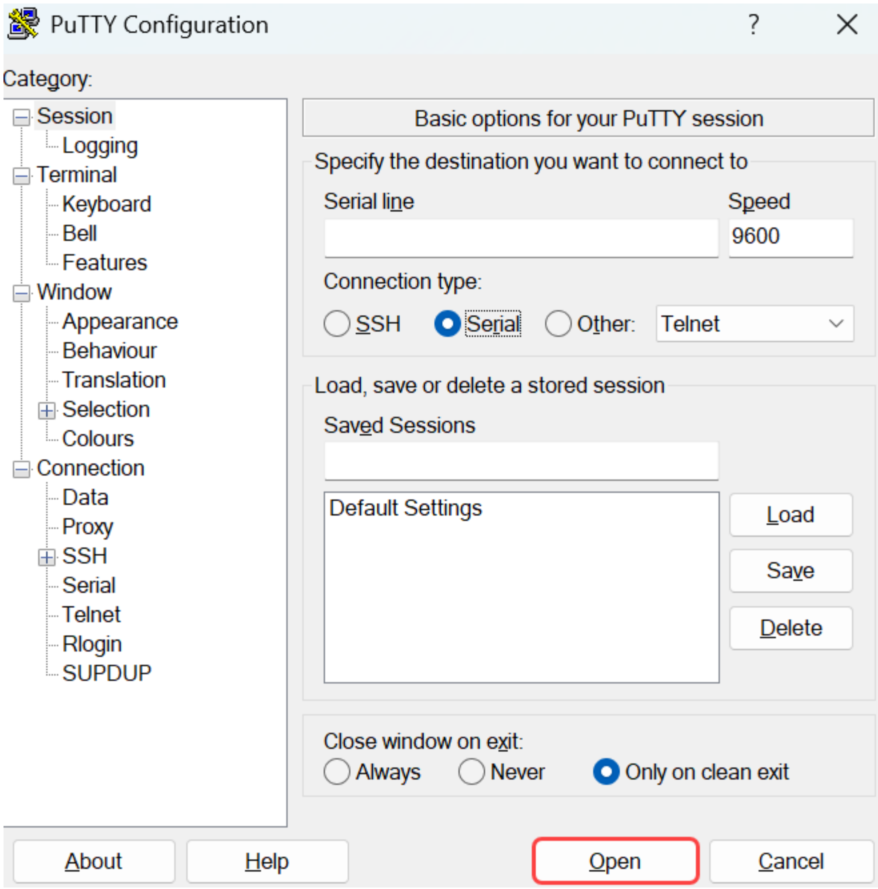
5. Type "PuTTY" into the windows icon that is on the taskbar and press the enter button. This should bring you to a window similar to the image above.



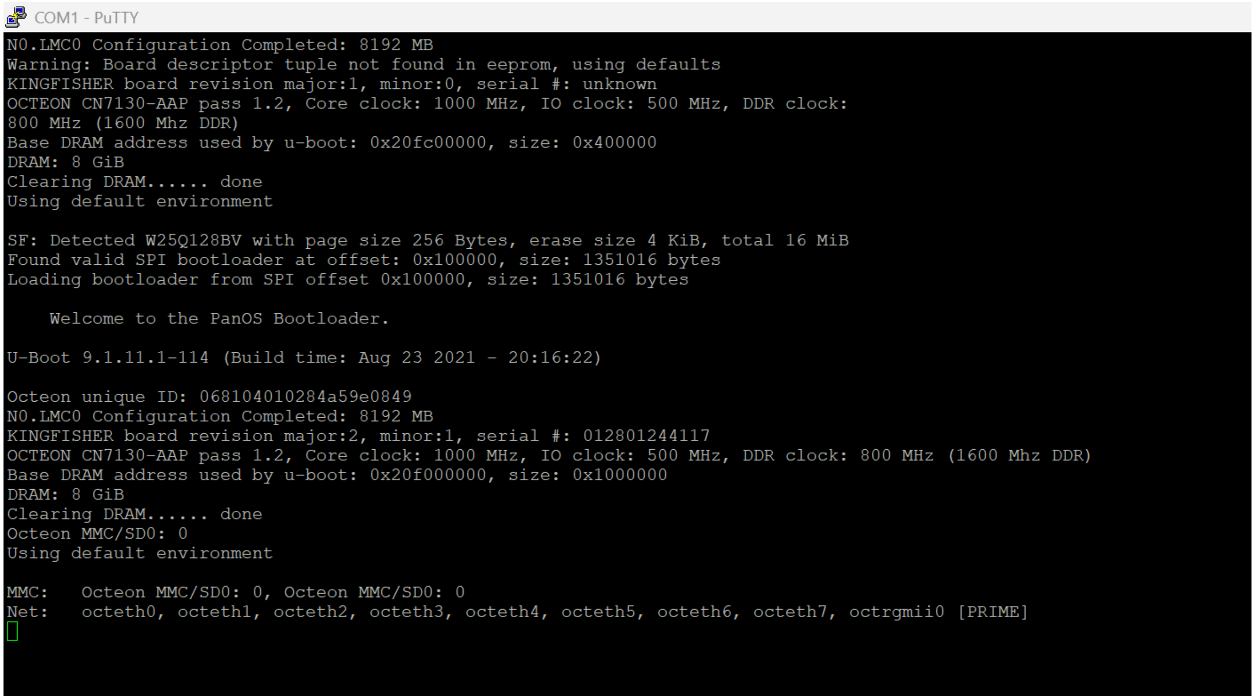
6. Press the Serial circle just as higlighted above.



7. Enter your COM port as noted before in a pervious step as highlighted above.

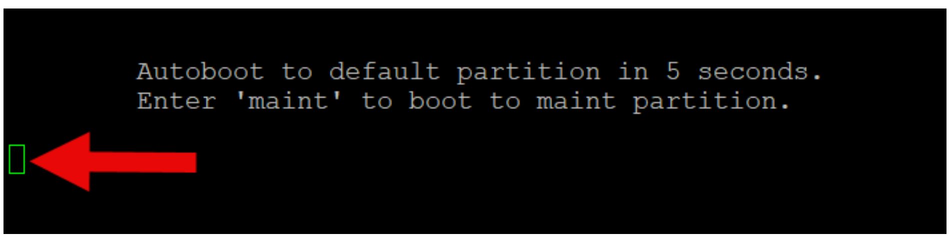


8. Select the Open button just as highlighted above.



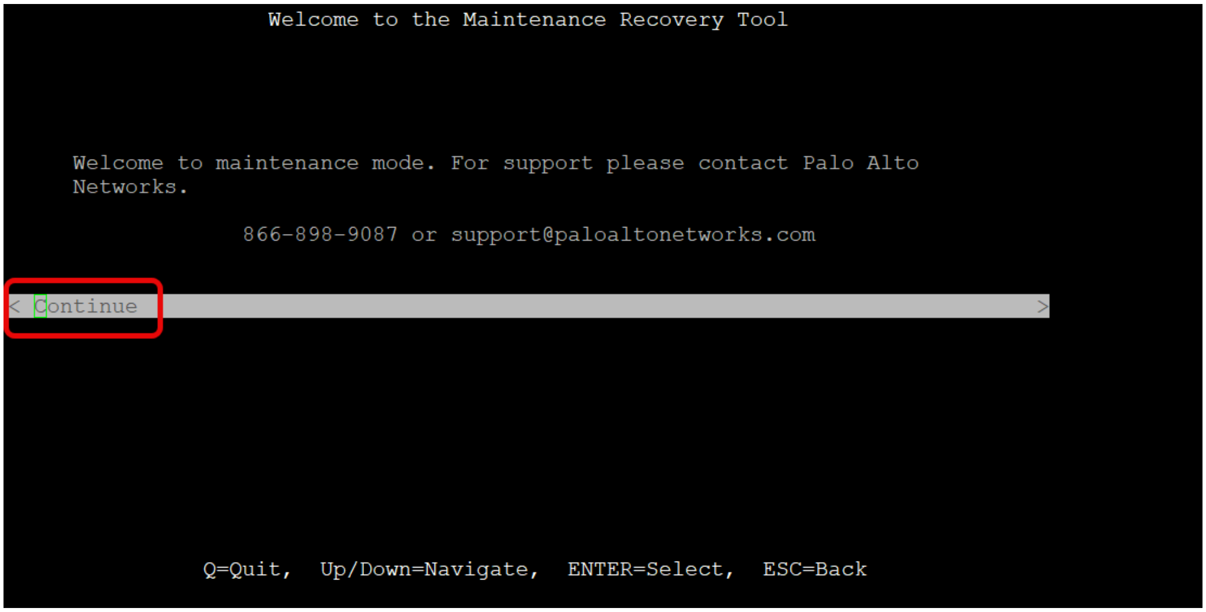
```
COM1 - PuTTY
N0.LMC0 Configuration Completed: 8192 MB
Warning: Board descriptor tuple not found in eeprom, using defaults
KINGFISHER board revision major:1, minor:0, serial #: unknown
OCTEON CN7130-AAP pass 1.2, Core clock: 1000 MHz, IO clock: 500 MHz, DDR clock:
800 MHz (1600 Mhz DDR)
Base DRAM address used by u-boot: 0x20fc00000, size: 0x400000
DRAM: 8 GiB
Clearing DRAM...... done
Using default environment

SF: Detected W25Q128BV with page size 256 Bytes, erase size 4 KiB, total 16 MiB
Found valid SPI bootloader at offset: 0x100000, size: 1351016 bytes
Loading bootloader from SPI offset 0x100000, size: 1351016 bytes

    Welcome to the PanOS Bootloader.

U-Boot 9.1.11.1-114 (Build time: Aug 23 2021 - 20:16:22)

Octeon unique ID: 068104010284a59e0849
N0.LMC0 Configuration Completed: 8192 MB
KINGFISHER board revision major:2, minor:1, serial #: 012801244117
OCTEON CN7130-AAP pass 1.2, Core clock: 1000 MHz, IO clock: 500 MHz, DDR clock: 800 MHz (1600 Mhz DDR)
Base DRAM address used by u-boot: 0x20f000000, size: 0x1000000
DRAM: 8 GiB
Clearing DRAM...... done
Octeon MMC/SD0: 0
Using default environment

MMC:   Octeon MMC/SD0: 0, Octeon MMC/SD0: 0
Net:   octeth0, octeth1, octeth2, octeth3, octeth4, octeth5, octeth6, octeth7, octrgmii0 [PRIME]
```

9. You should then be brought to a screen similar to the image above

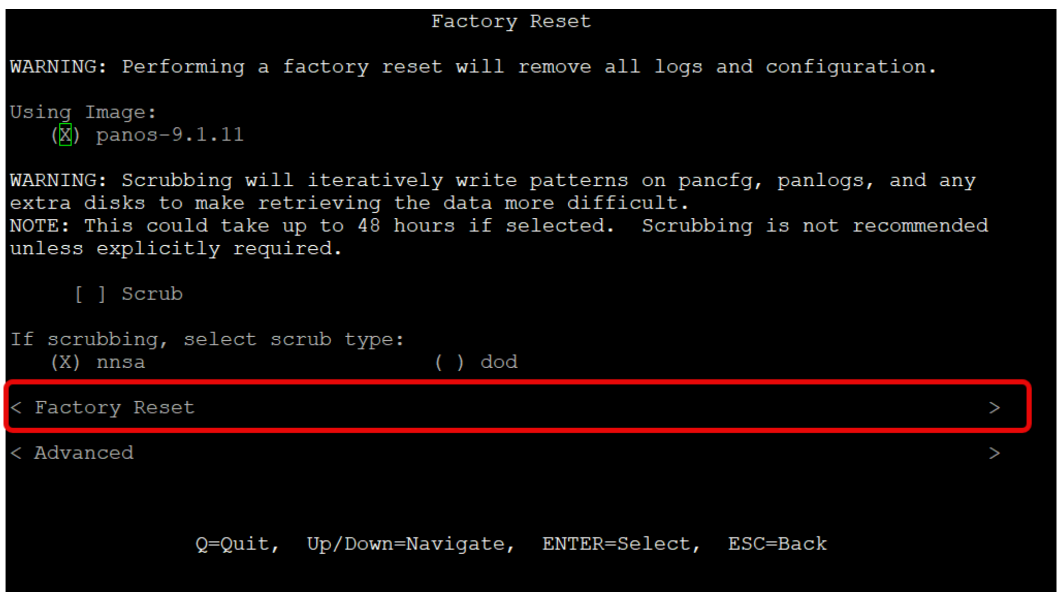


```
    Autoboot to default partition in 5 seconds.
    Enter 'maint' to boot to maint partition.
```

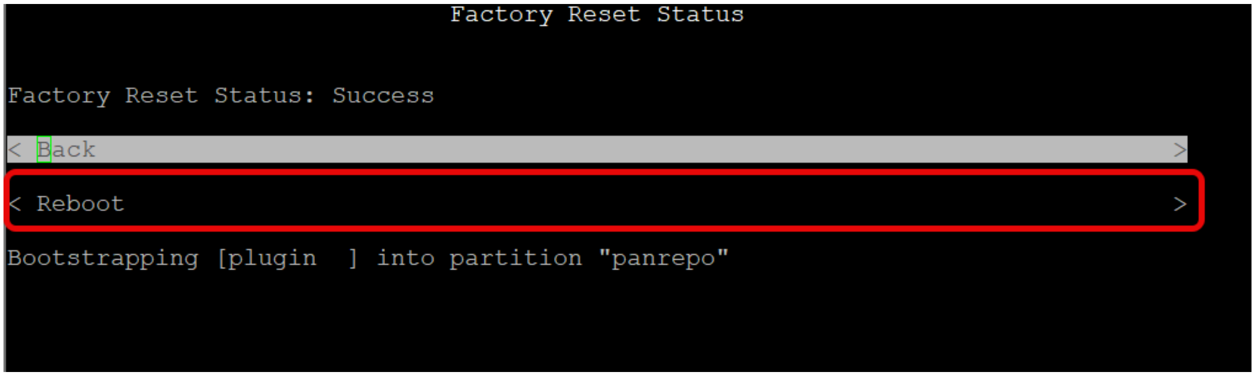
10. Wait at the screen until you get a message in text exactly like the image above and enter the wors “maint” within 5 seconds in the terminal as highlighted above.



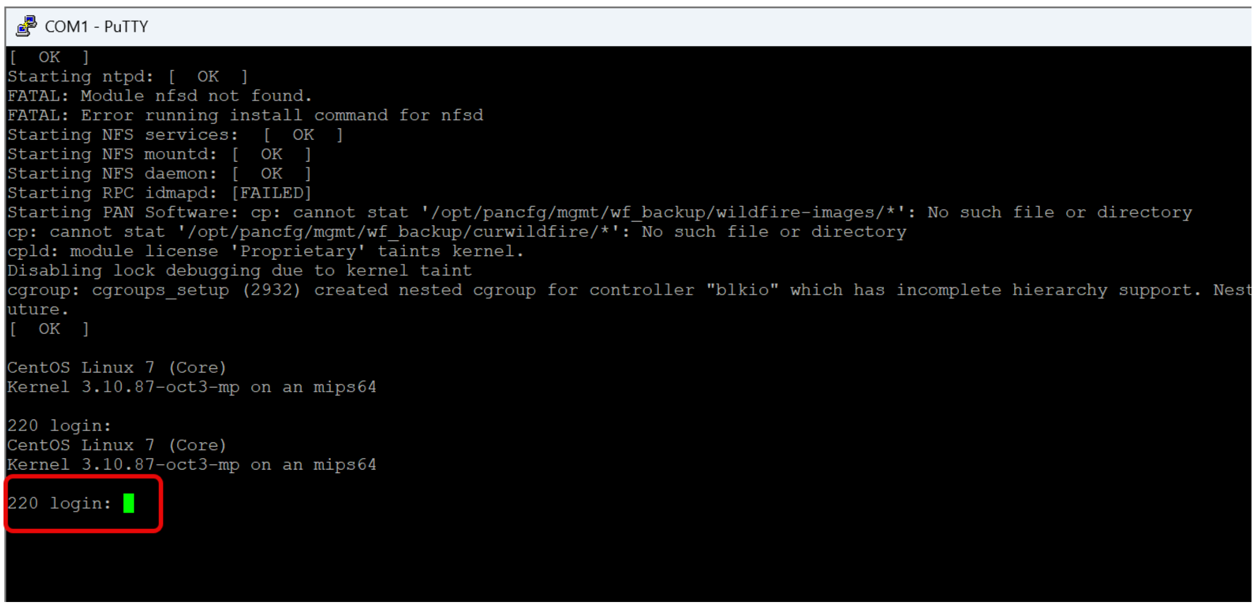
11. After waiting, you should then be brought to a screen as shown in the image above and then press the enter button as highlighted above.



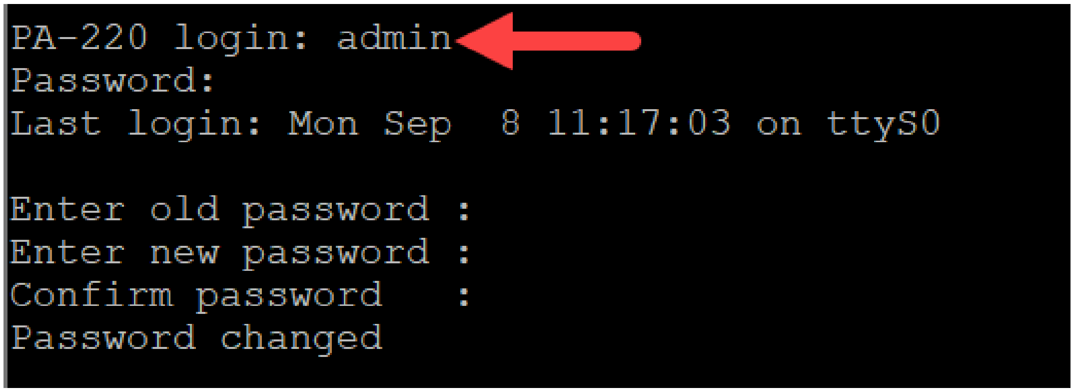
12. Press the down button until you get to the Factory Reset section as highlighted above and press the Enter button.



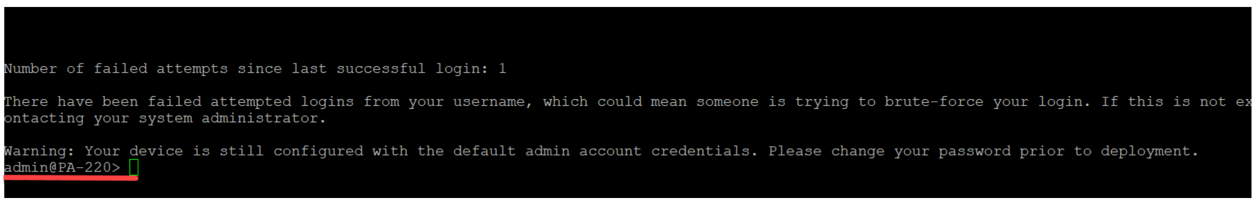
13. After waiting for the firewall to complete the Factory Reset and being brough to this screen, click the down arrow and press Enter on the Reboot section as highlighted above.



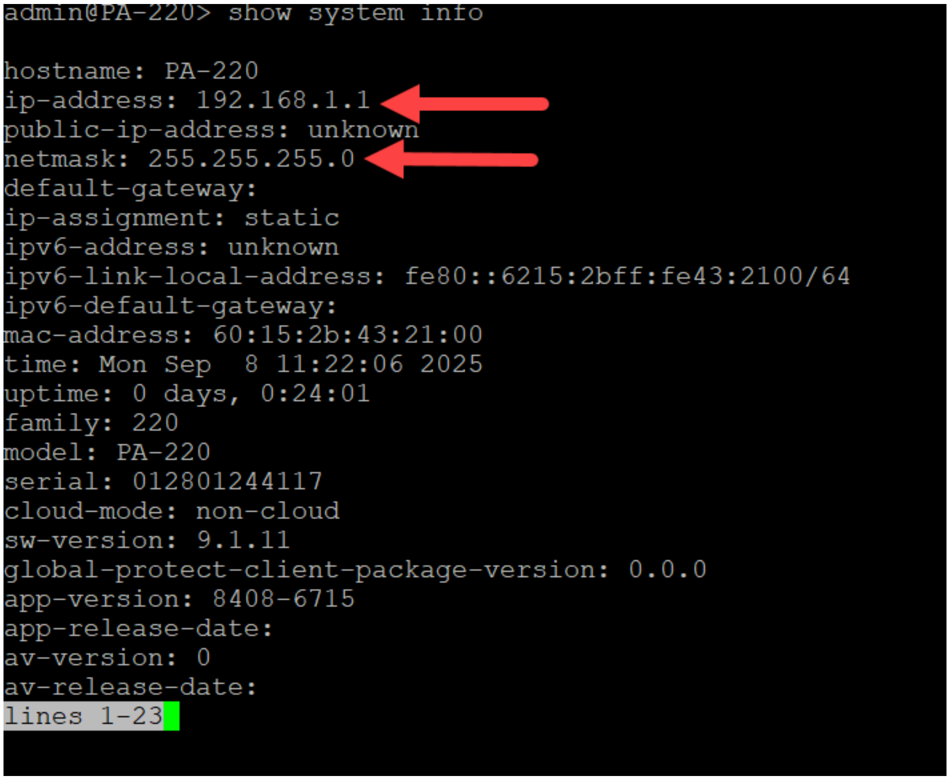
14. Then you should be brought to a screen like the image above and the terminal should say 220 login.



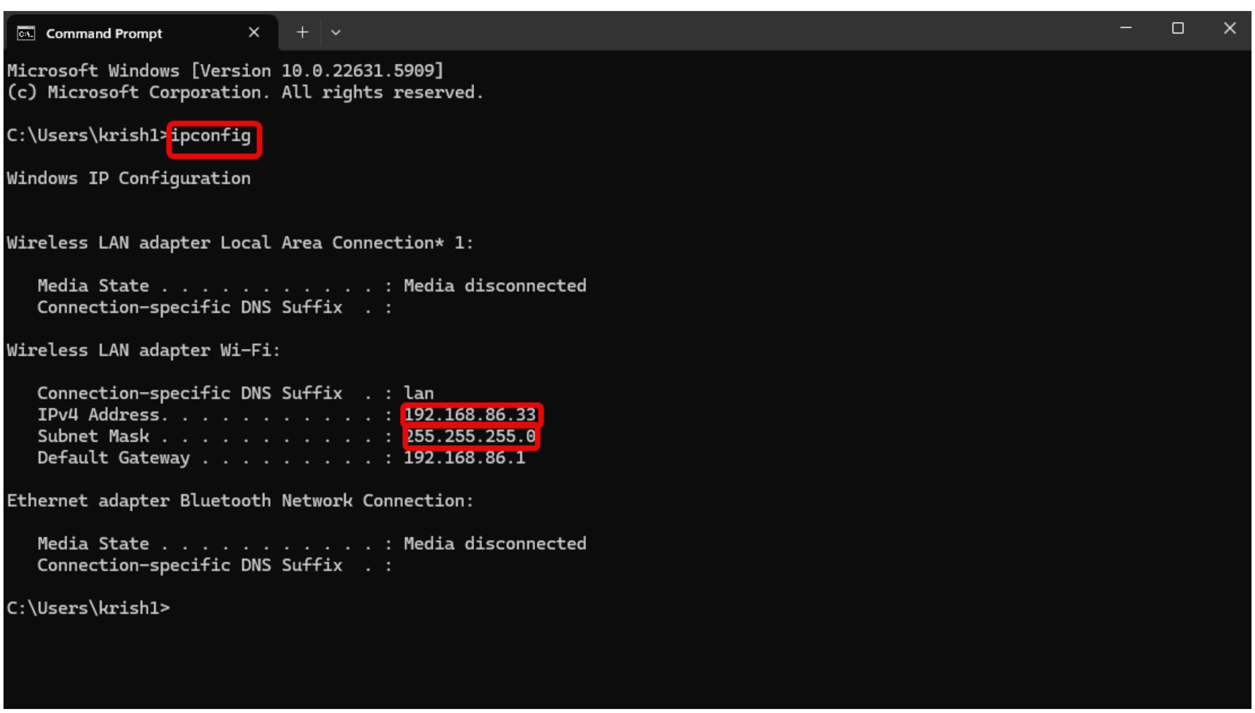
15. Type in a login username like admin as arrow indicates in the image above that you wish to save as the new configuration. Next type in a corresponding password. Type any string of words for the old password section. Then press the enter button, type in the password you wish to save for the firewall and type it again to confirm the configuration.



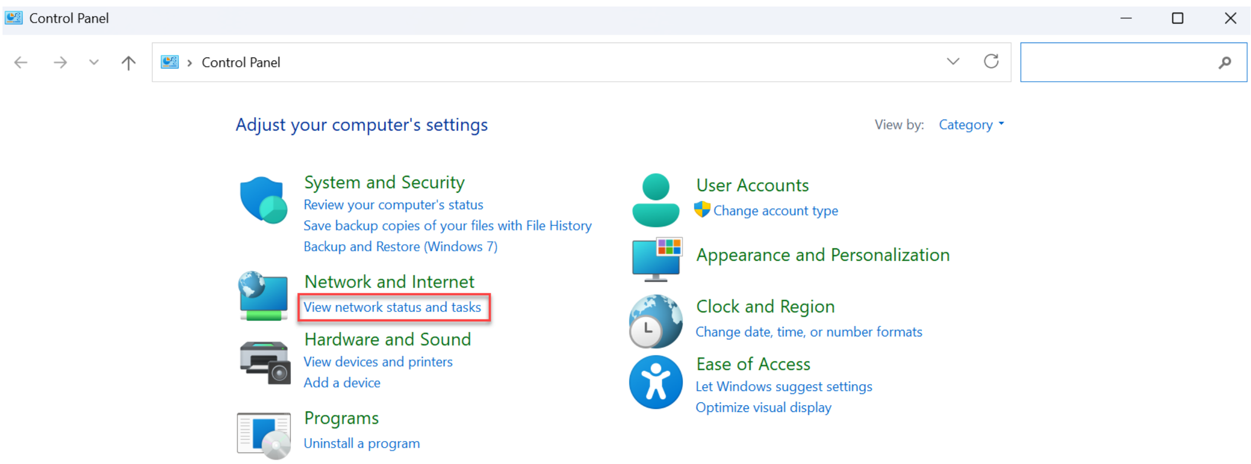
16. The firewall will then display text that indicates that the password is still not configured, and the previous password is still the account credential. However, you want to ignore this message and check that the terminal displays your username followed by @PA-220> to see if the configuration is temporarily saved. If so continue to the next steps, and if not unplug the firewall power cord and Factory Reset the firewall once again until you complete this step.



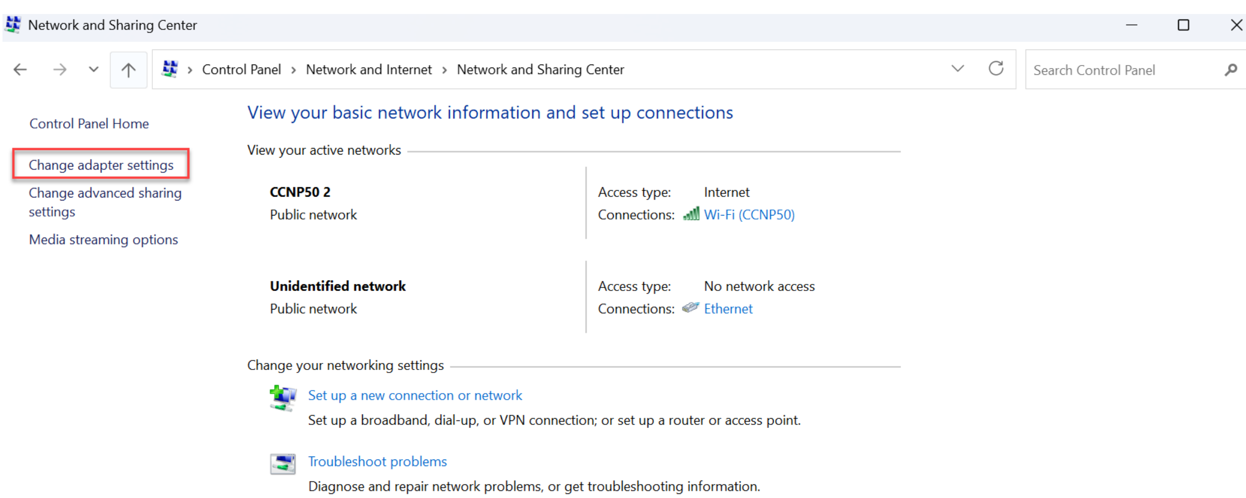
17. Then type in the command “show system info” as shown in the above image. Be sure to take note of the Ip address and netmask that should be in a similar format as shown by the arrows in the image above.



18. Type in Command prompt after clicking the Windows Icon and type the ipconfig command as highlighted in the picture above. Be sure to note down the IPv4 Address and the Subnet Mask as highlighted above.



19. Type in Control Panel after clicking on the Windows Icon in the taskbar at the bottom of the screen and after being brought to a window similar to the image above be sure to click the View network and status and tasks section as highlighted above.



20. After being brought to a windows similar to the image above, click the Change adapter settings as highlighted above.

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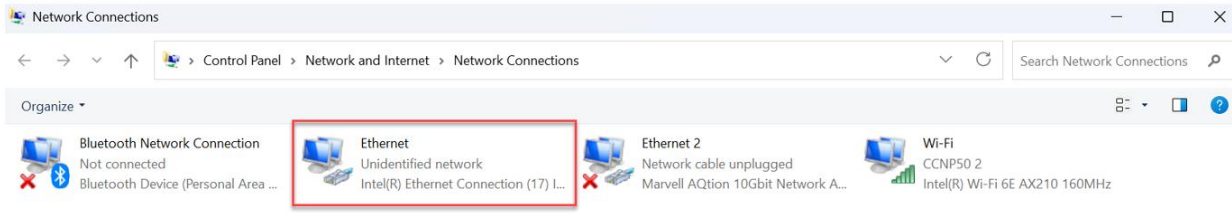
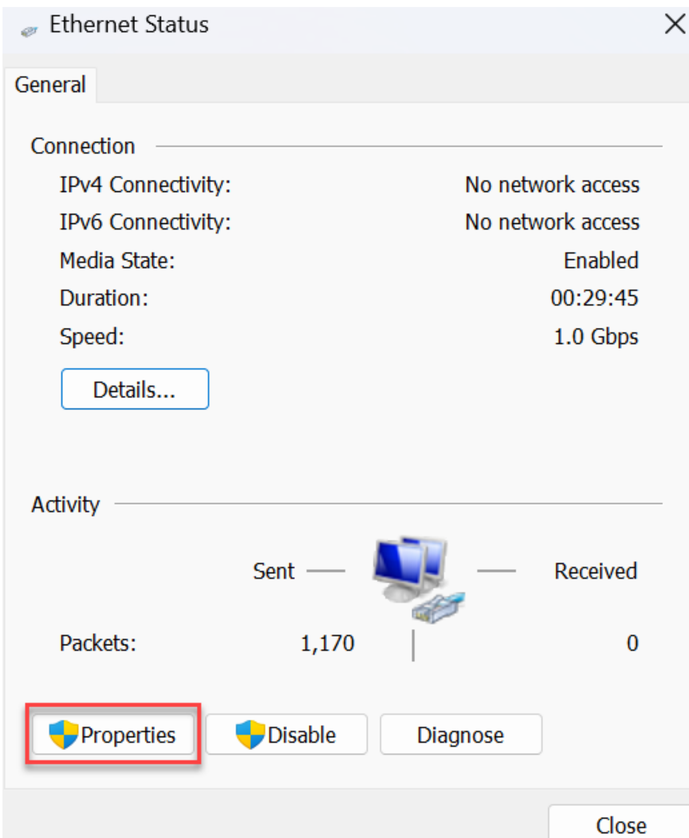
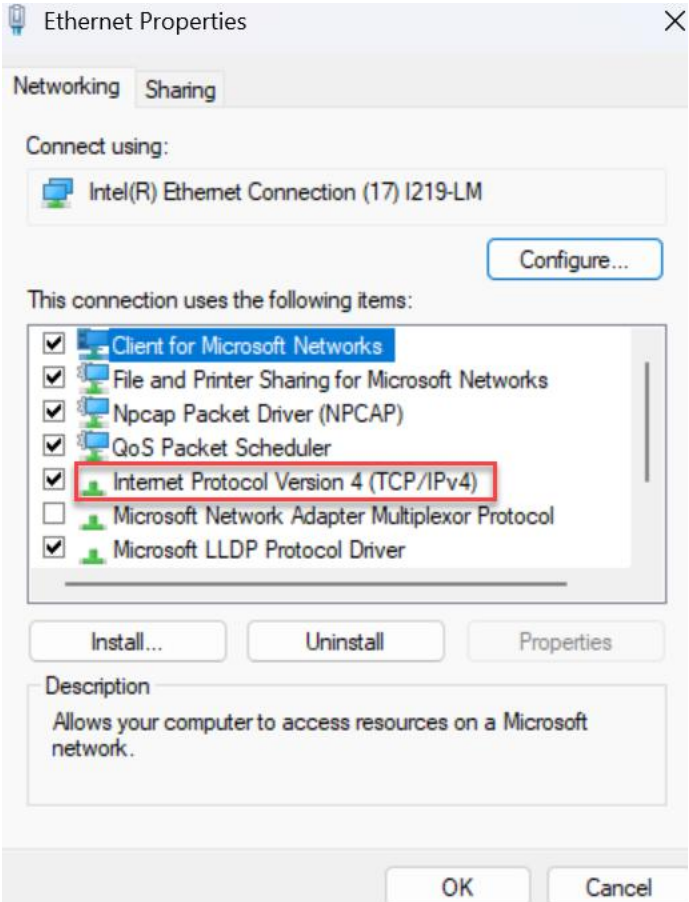


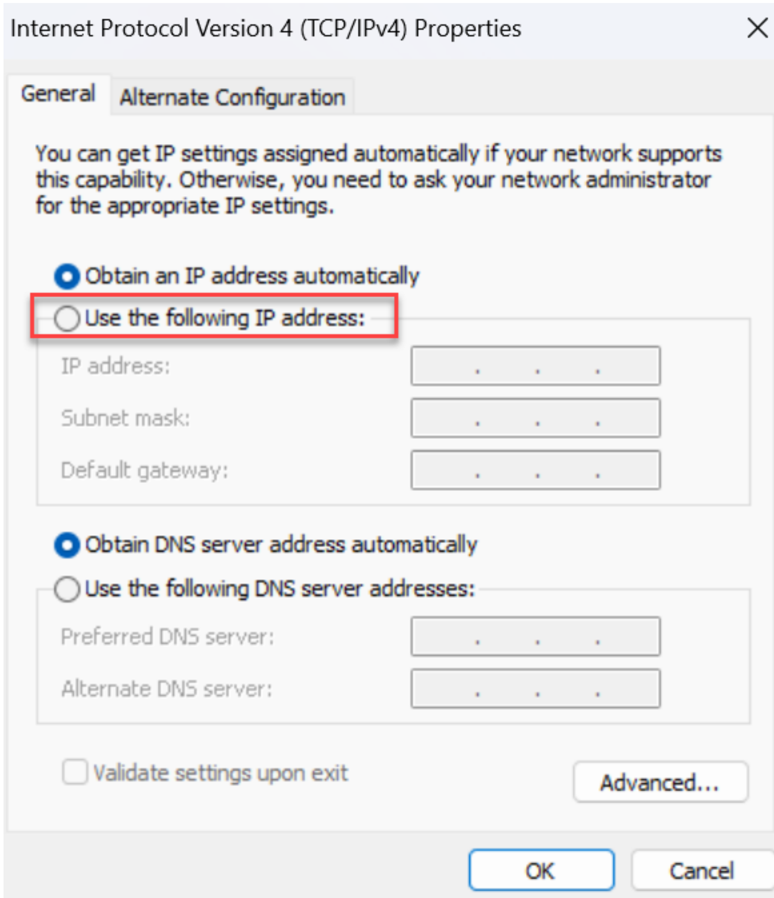
21. Click twice on the Ethernet Section as highlighted above.



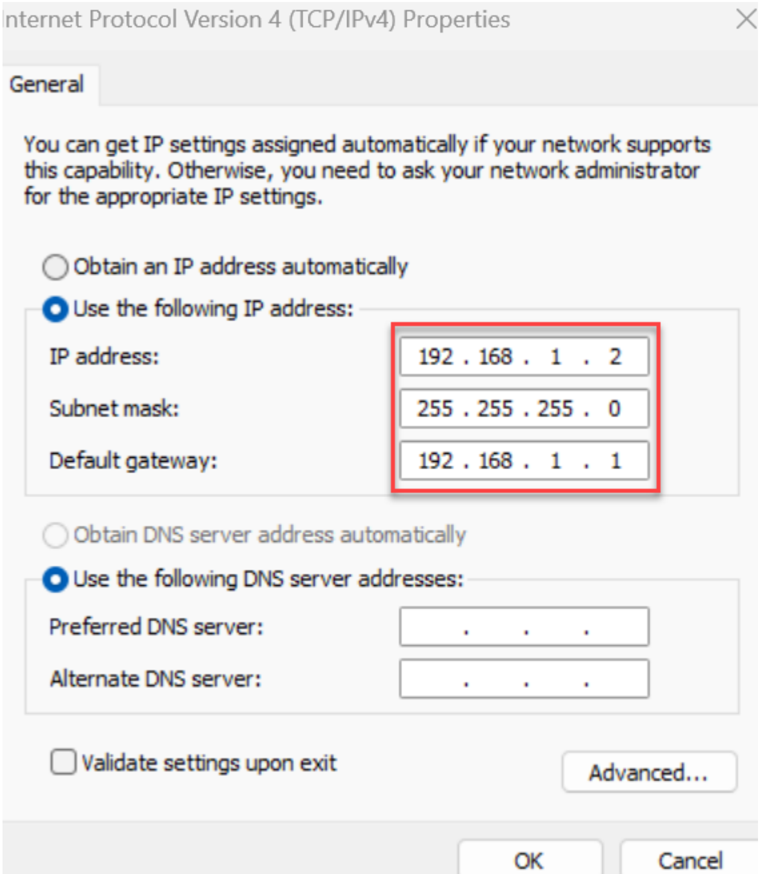
22. After being brought to an Ethernet Status page similar to the image above, click twice on the Properties button as highlighted above.



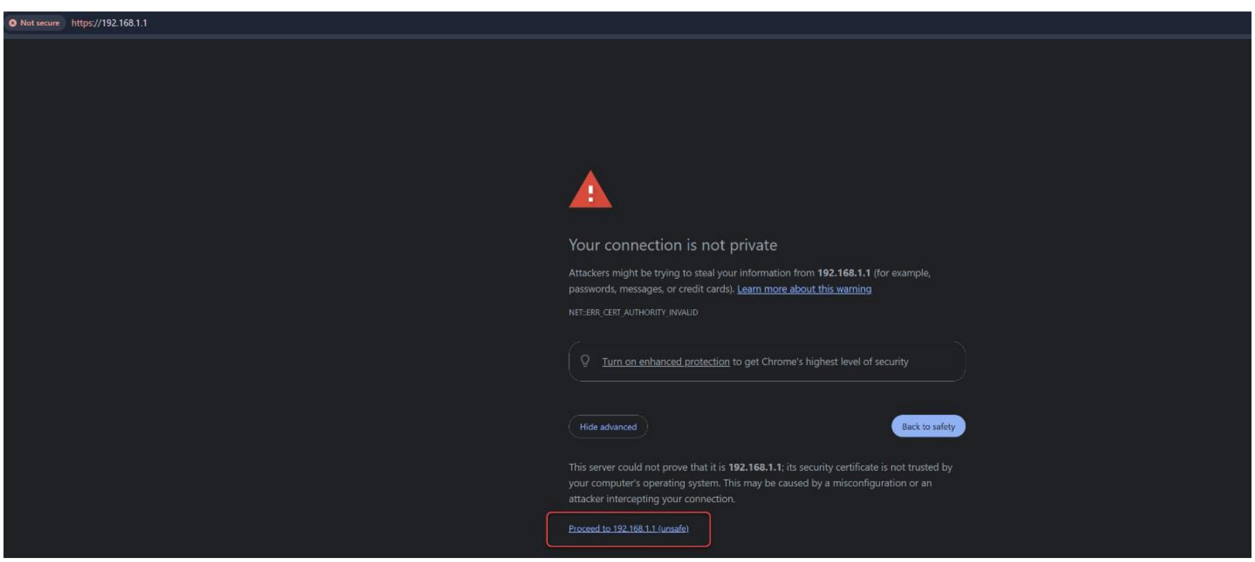
23. After opening the following window as shown in the image above, be sure to click twice on the Internet Protocol Version 4 (TCP/IPv4) as highlighted above.



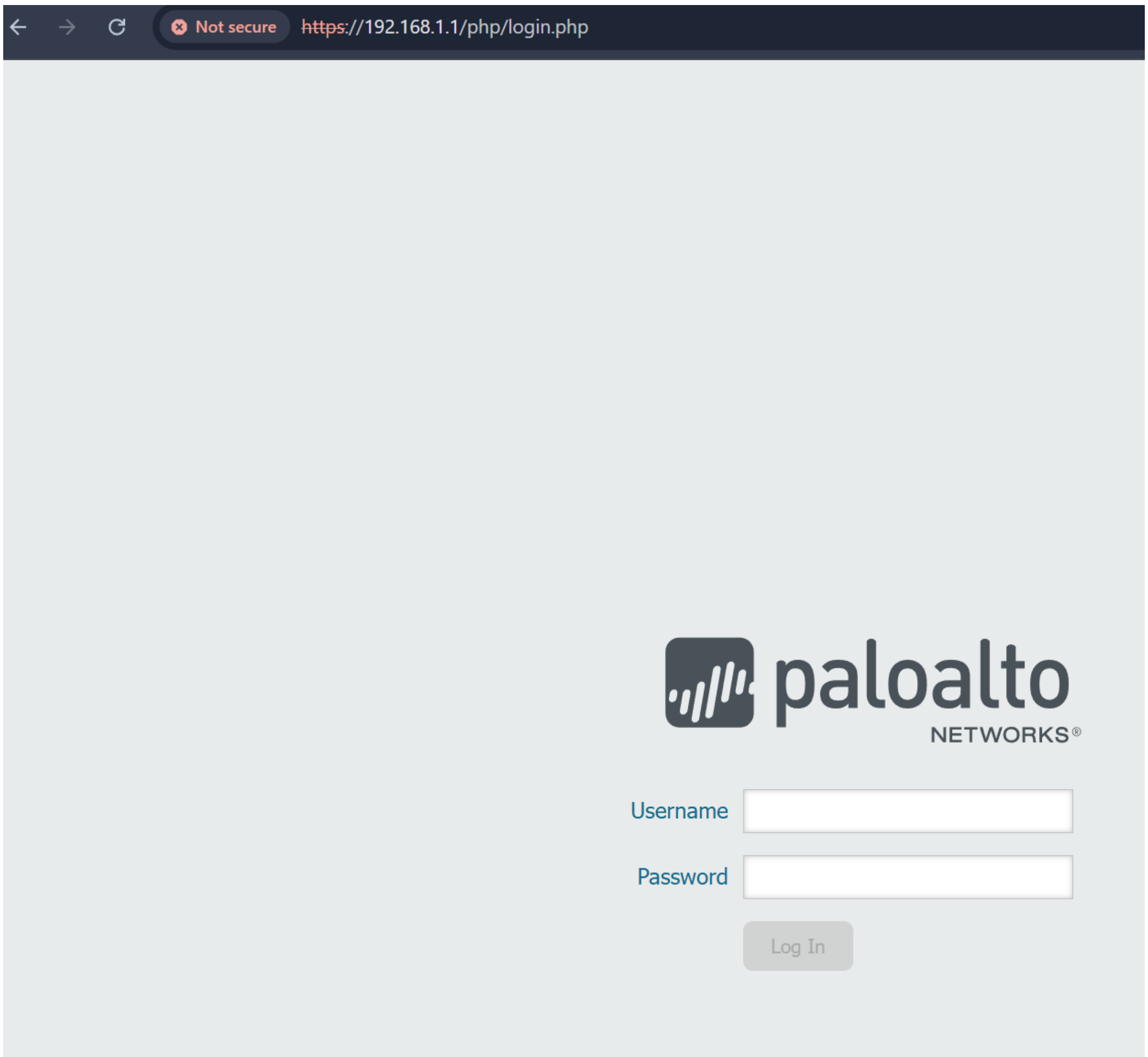
24. At this point you should see a window exactly like the image above and be sure to click the circle next to the Use the following IP address as highlighted above.



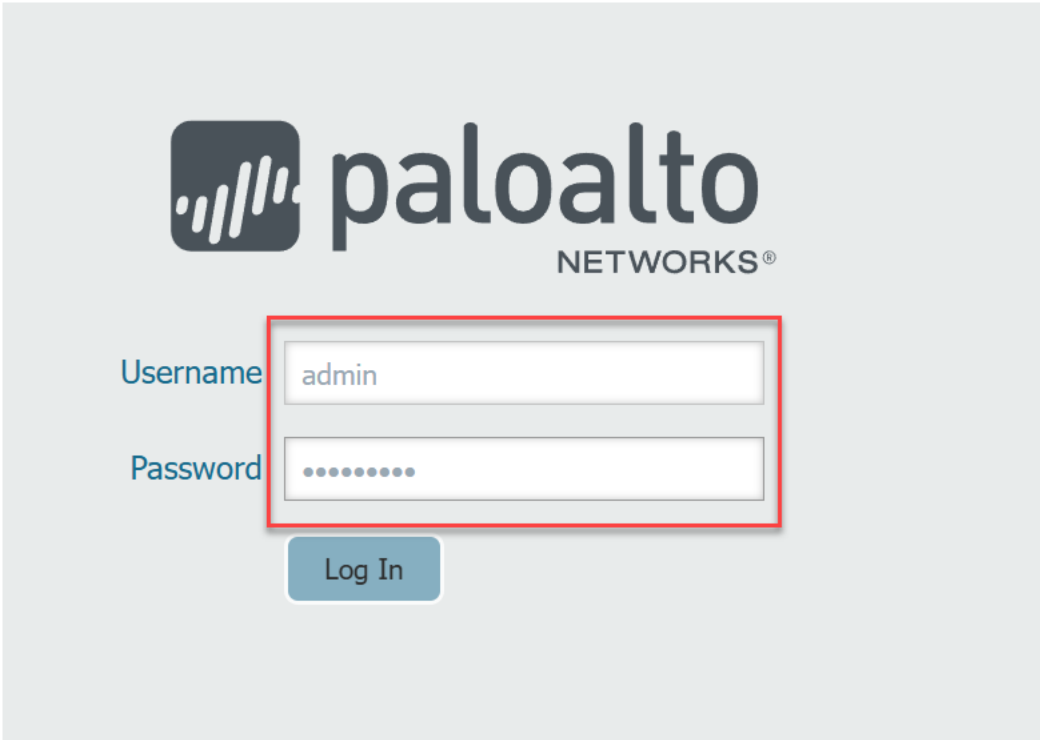
25. Type in the IP address and Subnet mask noted down from the Command prompt from a previous. For the Default Gateway be sure to write down the numbers that you noted down from the terminal of the Firewall.



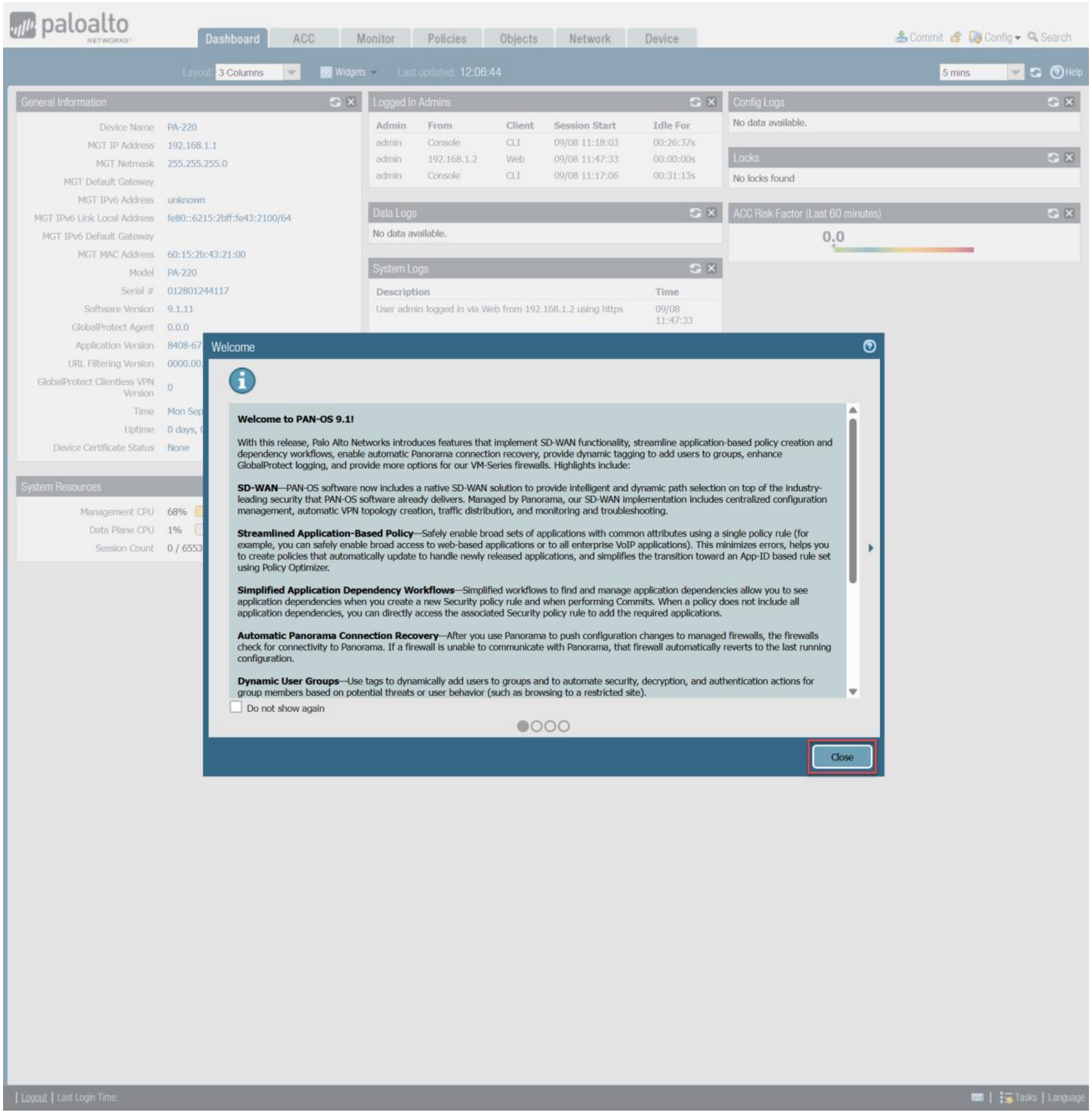
26. Go to your browser of choice and type https://________ with the underline being the IP Address of the Firewall that you noted down from the terminal of the Firewall from a previous step. Then click the show advanced button as highlighted above and then the Proceed to ____(unsafe) where the underline is the IP Address of the Firewall noted from a previous step.



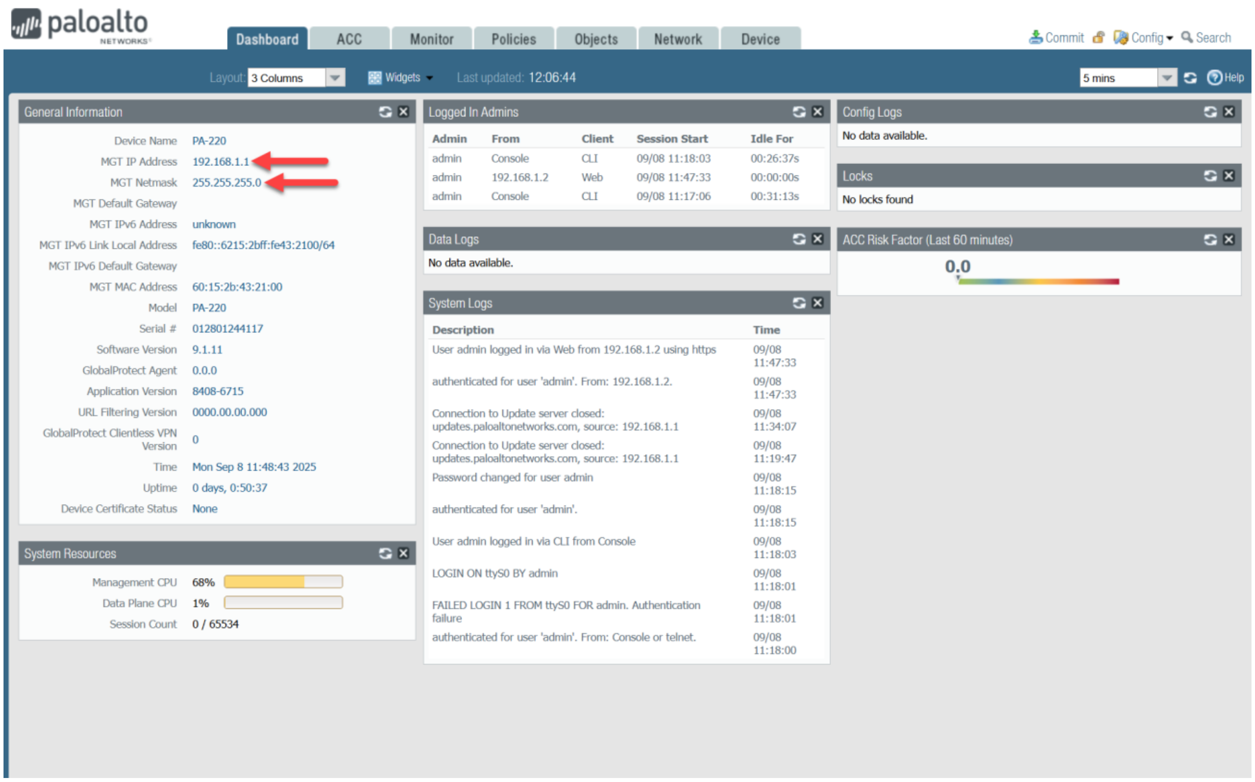
27. The window should then transition into a webpage similar to the image shown above.



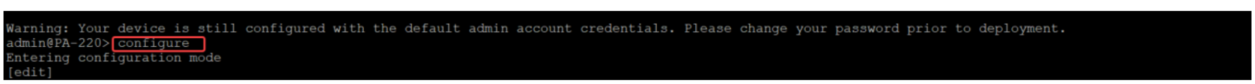
28. Enter in your username and password that you had temporarily set up from a previous step in the boxes as highlighted above. Then press the Log In button.



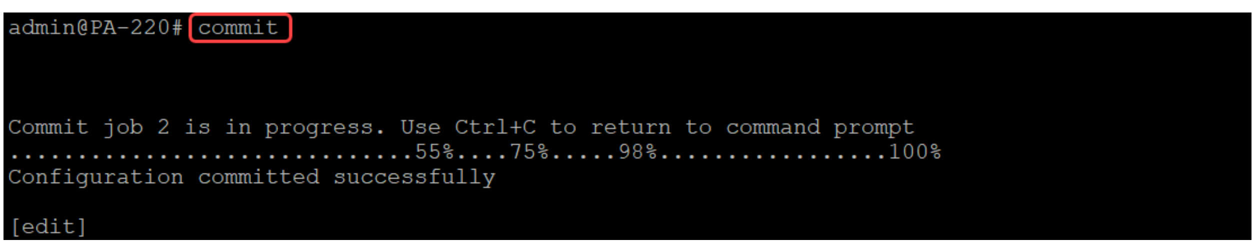
29. After being redirected to the next page a window should appear showing the current PAN-OS Software to which you should press the Close button as highlighted above.



30. In the Web GUI as shown above you will be able to see various details such as the Firewall’s IP Address, Netmask as indicated above, and many other sections with features that network administrators use to track, monitor and secure networks.



31. Go back to PuTTY and type the command configure as highlighted above.



32. Next, type in the command commit as highlighted above and the following text should be indicating how the “job” is being committed successfully.

```
admin@PA-220# set mgt-config users admin password
Enter password   :
Confirm password :

[edit]
admin@PA-220#
```

33. In a new line, type the command "set mgt-config users admin password" and type in the password you want to set for the firewall.

```
admin@PA-220# commit


Commit job 3 is in progress. Use Ctrl+C to return to command prompt
......................
```

34. As done in the previous step, type in the commit command and check if the "job" has committed successfully. Note: We commit the password again to make sure that for the few outlier cases where the password doesn't commit, we are completely accounted for.

Config Logs

Command	Path	Admin	Time
commit		admin	09/10 10:01:10
set	config mgt-config users admin	admin	09/10 10:00:09

35. Reopen the Web GUI, look at the homepage specifically the Config Logs section as highlighted above, then make sure that the commit and set command that was typed in the previous steps match the same timestamp in order to verify if the password was committed.

System Logs

Description	Time
Connection to Update server closed: updates.paloaltonetworks.com, source: 192.168.1.1	09/10 10:05:06
Commit job succeeded. Completion time=2025/09/10 10:03:03. JobId=3. User:admin	09/10 10:03:03

36. Then in the System Logs section check if the Commit job succeeded in the Description as highlighted above match the time stamp to finally verify that you have completed the authentication commits written in a previous step.

Problems

The 3 main problems that the personnel accessing this analysis could come across is the long wait time, confusing terminal statements, and misconceptions about how the new configurations that were typed are not saved even if the terminal states otherwise. In the beginning after plugging in the power cord the user will have to wait up to 20 mins after rebooting. Then after opening the terminal the user will have to wait up to 10 mins after waiting for maintenance mode to appear as an option. Even after Factory Resetting and Rebooting the user must be careful and wait for the 3rd light on the Firewall to turn from orange to green takes which takes approximately 15 minutes. Even when prompted to change a password and it seems that your password and new login credentials the configuration is not actually saved to the Firewall and won't be changed for the next time the Firewall is booted up which can be confusing for the user. The terminal will also show your failed attempts, that someone is brute forcing the Firewall which is you the user, and that your default admin account credentials is still not saved to the Firewall yet. In order to clear this misunderstanding, we included a section in the configurations on how to change your password properly and even how to see the change with a timestamp in the Web GUI as well as other basic functions that network administrators use to monitor traffic on a network.

Conclusion

In this analysis/guide, I created a step-by-step guide on how to essentially completely Factory Reset a PA-220 Next Generation Firewall as well as view the Web GUI interface of the Firewall. I then created a brief explanation on some of the sections that were essential in the Web GUI for network administrators. Before the main step by step guide, I gave an explanation on each of the relevant technologies, from the Pan-OS, Firewall, Cables, LAN vs. WAN, and the basics of PuTTY and its global usage. This analysis is enough for any personnel who decides they want to Factory Reset and access the Web GUI of a firewall to be able to even with minimal technical skills.